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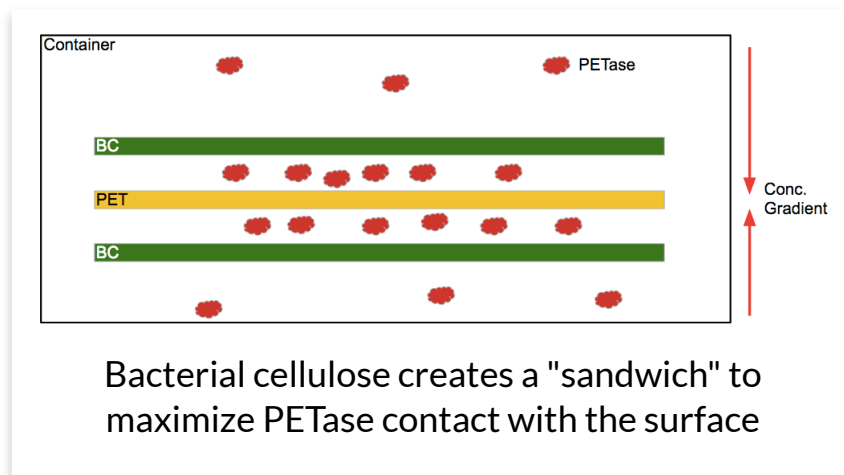
BACTERIAL CELLULOSE

Does the addition of a
cellulose binding domain
accelerate the degradation
of PET?

Model Setup

We considered the following degradation setup: a sheet of PET is “sandwiched” between two layers of bacterial cellulose, to which we add our PETase-CBD which is able to bind to the bacterial cellulose. The

bacterial cellulose, due to its insulating nature, will prevent PETase that is close to the PET film from escape into the rest of the solution, resulting in a high local concentration of PETase close to the surface.



We determined the surface area of a PET bottle modeling it as a perfect cylinder with diameter $d = 7$ cm, height $h = 23$ cm, and mass $m = 4$ g. which is about the average size of a disposable water bottle.

- Surface area of bottle = $2dh = (2)(7\pi \text{ cm})(23 \text{ cm}) = 322\pi \text{ cm}^2$

- Volume of PET = $w dh = (0.1 \text{ cm})(7\pi \text{ cm})(23 \text{ cm}) = 16.1\pi \text{ cm}^3$

We want to model the exact difference between the local and the global concentrations in such a scenario. In order to do this, we used an assumed value for the “working thickness” of

cellulose, which is the thickness at which making the cellulose layer thicker will yield a small increase in the insulating behavior of the cellulose layers. We used a value that is an underestimate of 0.1 mm as our starting value. We aim to use experimental data to adjust this value. We determined the following parameters:

- Cellulose density = 1.5 g/cm^3
- Total working weight of cellulose = $(1.5 \text{ g/cm}^3)(322\pi \text{ cm}^2)(0.01 \text{ cm}) = 15.87 \text{ g cellulose}$

Using values obtained from the literature about the CipA-CBD [1] where our CBD was acquired from, we determined:

- Maximum binding of CipA-CBD to cellulose = $0.43 \mu\text{M/g cellulose}$
- Local concentration of PETase at 50% binding = $(0.5)(15.87 \text{ g})(0.43 \mu\text{M/g}) = 3.41 \mu\text{M PETase}$

Enzyme Kinetics

The degradation of PET follows heterogeneous catalysis, where the reactants and the products [2] are in

different phases. This creates difficulty in modeling the rate of degradation but teams such as Tianjin China has used Langmuir's Equation, secretion, and diffusion models to model the process of secretion, diffusion, adsorption, and degradation. However, while their models gave great insight into the mathematical relationships between these variables, it was not possible to obtain practical results as many of the variables used in the equation are not found in the literature and are difficult to obtain experimentally.

We have developed the "working cellulose" model which is more practical in predicting the rate of degradation when PETase with a cellulose binding domain is added to a degradation system. We modeled the rate of degradation of PETase as a zero-order system, where the concentration of the reactant is negligible to the rate of the reaction. This is present in many surface-limited chemical catalysis models, where the catalyst surface is saturated by the reactants, and the rate limiting step can be simplified in the surface area of the catalyst.

However, in our PET degradation

model, the reverse holds true, where the concentration of PETase is the rate limiting variable compared to the surface area of the PET film when considering practical examples such as the degradation of a plastic bottle. We determined the following rate equation for the degradation of PET:

$$\frac{d[product]}{d\epsilon} = k[enzyme]dt \xrightarrow{\text{SolveDiffEq}} [product] = k[enzyme]t$$

In our reduced and simplified zero order enzyme kinetics equation, we only need to know the product, time, and enzyme concentrations to determine the kinetic variable k to predict degradation. Through values obtained through the original PETase discovery paper by Yoshida et al, we were able to calculate the kinetic variable for both low-crystalline and high-crystalline PET.

At 30°C, pH 7.0, [PETase concentration] = 50nM, time = 18 hrs: PETase produced a total of 0.015 mM of product when fed high-crystalline PET and 0.3 mM of product when fed thin-sheet PET.

Using above data we can determine

the rate constants.

$$k_{hc} = \frac{0.015 \cdot 10^{-3} M}{5 \cdot 10^{-8} M \cdot 64800 s} = 0.00463 s^{-1}$$

$$k_{lc} = \frac{0.30 \cdot 10^{-3} M}{5 \cdot 10^{-8} M \cdot 64800 s} = 0.0926 s^{-1}$$

Results

Therefore, with the calculated local concentration of our enzyme and our enzyme kinetic parameter we were able to predict the rate of degradation of a plastic water bottle. If we desire 4 grams of PET to completely degrade to its subsequent product MHET, using the molecular weight of MHET ($M_w = 210.185 \text{ g mol}^{-1}$), we determine that the final concentration of MHET in a 1 L container would be 0.019 M. Since PET is not an aqueous substance, it is difficult to assign it a "concentration" value for our equation. Therefore we would determine the creation time of MHET rather than the degradation time of PET.

Using our equation:

$$[product] = k[enzyme]t$$

With the following parameters:

- [PETase] = 3.41 M
- $k = 0.00463 \text{ s}^{-1}$ for high crystalline PET
- [MHET] = 0.019 M

Solving for time, we predict that complete degradation of one PET bottle sandwiched between two bacterial cellulose sheets with 50% saturation of our PETase-CBD fusion protein will occur in 1.2×10^6 seconds, or approximately 14 days.

As degradation progresses, the appropriateness of our zero-order enzyme kinetics model will decrease as the surface area of PET available to our enzyme will become more significant. Therefore, a time to 50% degradation will be more indicative of the actual degradation of PET. We determined this value to be about 7 days.

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- 1: Sabathé, F. & Soucaille, P. Characterization of the CipA scaffolding protein and in vivo production of a minicellulosome in *Clostridium acetobutylicum*. J. Bacteriol. 185, 1092–1096 (2003).
 - 2: Yoshida, S. et al. A bacterium that degrades and assimilates poly(ethylene terephthalate). Science 351, 1196–1199 (2016).

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